Advanced Financial Modeling, Module 3: More Advanced Accounting (Reference Module)

You'll start this module by learning about the Book Value, Face Value, and Market Value of Debt, and how they differ because of items such as Issuance Fees and Original Issue Discount (OID). You'll also learn about different types of Debt, non-cash or Paid-in-Kind (PIK) Interest, and convertible bond accounting.

Then, you'll learn about Equity Investments and Noncontrolling Interests, see how they get created and flow through the financial statements, and see what happens when they disappear as companies deconsolidate their statements.

You'll wrap up with a few lessons on more specific topics, such as the tax treatment of Stock-Based Compensation, Unrealized Gains and Losses on different types of securities, FIFO vs. LIFO for Inventory, and introductory Pension Accounting for defined-benefit pension plans.

AFM-03-01: Debt Issuance Fees and the Book Value vs. Face Value of Debt (20:41)

In this lesson, you'll learn the difference between the Book Value and Face Value of Debt, and you'll learn how to amortize Debt issuance fees and reflect the proper accounting treatment when an entire Debt issuance is repaid before its maturity date.

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- **2:45:** Book Value vs. Face Value and Debt Issuance Fees
- **5:58:** Initial Issuance and Repayment Upon Maturity
- **11:52:** Early Repayment of Debt
- **17:37:** Deferred Taxes
- **18:42:** Recap and Summary

Quiz Questions:

11:51 When do the Book Value and Face Value of a Debt issuance differ the MOST?

a) **When the Debt is first issued.**
b) Upon maturity, when the Debt is repaid.
c) It depends on the Amortization schedule for the issuance fees.
d) It depends on whether or not early principal repayments are allowed.

Explanation: The first answer choice is correct. The Book Value and Face Value of Debt always differ the most when the Debt is FIRST ISSUED because at that point, Book Value = Face Value – Issuance Fees. Each year after that, the difference narrows as more of the Issuance Fees are amortized. The second answer is completely wrong because Book Value = Face Value upon

maturity, and this question has nothing to do with the Amortization schedule or early principal repayments (they don't affect the percentage difference – just the amount of remaining fees to be amortized).

AFM-03-02: Market Value of Debt (11:41)

You'll learn about the Market Value of Debt in this lesson, why it might differ from the Book Value of Debt, and how it might influence a company's financial statements – even though changes in the Market Value do not make a direct impact.

Table of Contents:

- **0:37:** The Market Value of Debt
- **2:44:** Excel Setup and Calculations
- **8:52:** So… Why Does This Matter?
- **10:29:** Recap and Summary

Quiz Questions:

8:52 If prevailing yields on similar Debt Issuances are initially 5%, the company issues Debt with a coupon rate of 5%, and then prevailing yields rise to 7%, what happens to the Market Value of this company's Debt?

a) It should increase because investors can now earn a higher yield.
b) It should decrease because this Debt is now comparatively less attractive than other issuances.
c) We need additional information about the company's credit quality and the maturity of the Debt to answer this question.

Explanation: The second answer choice is correct. When the Discount Rate goes up, the Present Value of any security, including a bond (Debt), decreases because it means that there are other, higher-yielding options in the market. The first answer makes no sense, and the third answer is partially true because if the credit quality or maturity changes, those could also affect the Market Value. However, we don't *need* that information to answer the question – if they stay the same, we can predict the change in Market Value using only the change in prevailing yields.

AFM-03-03: Debt Discounts, Premiums, and Original Issue Discount (OID) (20:58)

In this lesson, you'll learn how to reflect an Original Issue Discount (OID) on the financial statements, and how it changes with early Debt principal repayment and unusual amortization schedules; you'll also make the formulas for Debt Issuance Fees more robust.

Table of Contents:

- **0:53:** Introduction
- **3:10:** OID Without Debt Repayments
- **8:58:** OID With Debt Repayments
- **14:51:** Generalizing the Issuance Fee Formulas
- **18:47:** Recap and Summary

Quiz Questions:

14:50 If a company issues $100 of Debt with a 10-year maturity and an OID of $5, and it then repays $0 of the principal in Year 1 and all $100 in Year 2, what happens to the OID in Years 1 and 2?

a) It decreases by $0.0 in Year 1, and then the full $5.0 is amortized in Year 2.
b) It decreases by $0.5 in Year 1 and by $0.5 in Year 2, following the normal straight-line schedule.
c) It decreases by $0.5 in Year 1, and then the full remaining $4.5 is amortized in Year 2.
d) None of the above.

Explanation: The third answer choice is correct. When there is no Debt principal repayment, OID simply amortizes on a straight-line schedule, at $5 / 10 = $0.5 per year in this case. That rules out answer choice #1. When the company repays Debt principal, then it reduces the OID balance proportionally to the amount of Debt principal repaid relative to the initial Face Value of the Debt. So, if it is 50% repayment, the OID will fall by 50%, split into Amortization and "Loss" components (the Loss just represents accelerated Amortization). When it's 100%, as in this question, the full amount of remaining OID is "amortized" in Year 2 (with $0.5 of actual Amortization and $4.0 recorded as a "Loss").

AFM-03-04: Introduction to Convertible Bond Accounting (23:10)

You'll learn how and why companies issue Convertible Bonds in this lesson, and you'll understand how the accounting works from the initial issuance to the Amortization of the Debt Discount and the eventual conversion or maturity of the Bond.

Table of Contents:

- **0:59:** Convertible Bond Accounting
- **2:42:** Step 1: Initial Convertible Bond Issuance
- **5:21:** Step 2: Handling Maturities and Conversions
- **8:15:** Step 3: Amortization of the Debt Discount
- **12:37:** Step 4: Linking the Statements
- **21:30:** Step 5: Recap and Summary

Quiz Questions:

8:14 When forecasting the Conversions line item for a Convertible Bond, why is it convenient to use the MIN function with the Beginning Face Value of the Bond?

a) **To handle the case where the maturity happens before the conversion.**
b) To handle the case where part of the Convertible Bond principal, but not all of it, is repaid early.
c) To handle the case where the company repurchases the Bond early at a premium.
d) All of the above.

Explanation: The first answer choice is correct. Since the user of our model could set the Maturity Year and the Conversion Year to any numbers, we need to check for the case where the conversion is in Year 5, but the maturity is in Year 3 (for example) and ensure that the Face Value only goes to $0 once. To do that, it's convenient to use MIN to check the potential conversion in a year against the beginning Face Value in that year. The second answer is wrong because early partial repayment is almost unheard of for Convertible Bonds, and we're not handling the third scenario in this model.

21:30 What is the PRIMARY difference in the accounting treatments of Original Issue Discount (OID) on a Traditional Bond vs. the Discount on a Convertible Bond?

a) The tax deductibility of the Amortization of OID and the Convertible Bond Discount differ.
b) The Convertible Bond Discount does not amortize over time, but the Original Issue Discount does.
c) **When a Convertible Bond converts into shares, both its components are transferred into Common Shareholders' Equity.**
d) Book Value and Face Value tend to be closer for a Traditional Bond.

Explanation: The third answer choice is correct. When a Traditional Bond with OID is repaid early, the remaining Unamortized OID shows up as a Loss on the Income Statement, and the Book Value and Face Value both go to 0. But with a Convertible Bond, an early conversion

results in both the Liability and Equity components being transferred into Common Shareholders' Equity. The tax-deductibility of these items does not differ, and both amortize over time, so the first two answers are wrong. The last answer is wrong because Book Value and Face Value are not necessarily any closer for a Traditional Bond – it depends on the interest rates and credit quality of the company.

AFM-03-05: Types of Debt and Paid-in-Kind Interest (PIK) (11:11)

In this lesson, you'll learn about different types of Debt, including the Secured vs. Unsecured categories, and you'll understand how non-cash, accrued, or paid-in-kind (PIK) interest flows through the financial statements.

Table of Contents:

- **0:39:** Types of Debt
- **3:48:** Mechanics of PIK Interest
- **7:06:** Financial Statement Links
- **9:35:** Recap and Summary

Quiz Questions:

9:34 When interest on Debt is Paid-in-Kind (PIK) for one period, what happens to the Debt's Book Value and Face Value?

a) Book Value increases, but Face Value stays the same.
b) They both increase.
c) Face Value increases, but Book Value stays the same.
d) Neither Face Value nor Book Value increase.

Explanation: The second answer choice is correct. PIK Interest increases both the Book Value and Face Value of the Debt because it means the company will have to pay interest on a higher principal in the future, and it also means that the corresponding Liability line item on the Balance Sheet will increase, even more than it already increases from the Amortization of issuance fees and OID.

AFM-03-06: Book Value vs. Market Value vs. Face Value of Bonds: How to Keep Them All Straight (19:13)

You'll get a summary of the key Debt-related points from the first five lessons in this module here, and you'll learn how to apply Face Value, Book Value, and Market Value to real Debt schedules for Toro and Atlassian.

Table of Contents:

- **0:55:** Short Summary
- **3:11:** Toro Debt Schedule
- **8:05:** Atlassian Debt Schedule
- **17:27:** Recap and Summary

Quiz Questions:

8:04 How can you SIMPLIFY a Debt Schedule for a company that is continuously issuing and repaying Debt if you want to reflect the Book Value and Face Value each year?

a) Assume that the average maturity for each Debt issuance is beyond the end of the projection period.
b) Do not track the Issuance Fees as a separate line item; just the Amortization and initial deduction for them.
c) Ignore the possibility of accelerated Issuance Fee amortization upon early repayment.
d) All of the above.

Explanation: Everything above is correct. Debt Schedules are much simpler if the maturities and full repayments do not happen within the schedule (answer #1) because then you do not have to handle accelerated Issuance Fee or OID Amortization (answer #3). Also, you can save time and space by not tracking Issuance Fees separately; just deduct the initial fees from the Book Value and add the Amortization of those fees in each period.

AFM-03-07: Equity Method of Accounting (Equity Investments or Associate Companies), Part 1: Net Income and Dividends (15:54)

In this lesson, you'll learn what happens on the three financial statements when a company first purchases a minority stake (Equity Investment) in another company, and you'll see how the other company's Net Income and Dividends flow through the statements.

Table of Contents:

- **1:01:** Equity Method of Accounting ("Equity Investments" or "Associate Companies")
- **5:50:** Part 1: Initial Purchase of a Minority Stake
- **10:22:** Part 2: Net Income and Dividends from Equity Investments

- **13:51:** Recap and Summary

Quiz Questions:

13:50 A Parent Company purchases a 40% stake in another company for $40 at the end of Year 1. The other company generates $10 in Net Income and issues $5 in Dividends in Year 2. What is the Equity Investments line item on the Parent Company's Balance Sheet at the end of Year 2?

a) $45.
b) $35.
c) $38.
d) $42.
e) Cannot answer this question without knowing the Parent Company's Tax Rate.

Explanation: The fourth answer choice is correct. The initial purchase creates an Equity Investment of $40. Then, the Parent records 40% * $10 = $4 of Net Income from the Associate on its Income Statement. It then reverses this $4 on the CFS and records 40% * $5 = $2 for Dividends. The $4 adds to Equity Investments on the Balance Sheet, and the $2 reduces it. $40 + $4 – $2 = $42. The Tax Rate of the Parent does not matter at all, so the last answer choice is wrong.

AFM-03-08: Equity Method of Accounting (Equity Investments or Associate Companies), Part 2: Changing the Ownership and Gains and Losses (11:50)

You'll learn how to record changes in minority stakes (Equity Investments) in this lesson, including how Realized Gains and Losses work, and why determining the cost basis can be tricky.

Table of Contents:

- **0:56:** Increasing the Stake
- **2:15:** Sales and Realized Gains and Losses
- **7:56:** Testing the Model
- **9:54:** Recap and Summary

Quiz Questions:

9:53 Company A has a 30% stake in Company B in Year 2 when 100% of Company B is worth $200. Then, in Year 2, Company B's value increases to $250, and Company A sells its entire stake. Why is the Gain NOT necessarily equal to $15?

a) Because Company A might have acquired portions of its stake in Company B at different valuations.
b) Because Company A's cost basis in Company B might have been affected by Net Income and Dividends from Company B.
c) Because Company A might have previously sold portions of its stake in Company B at different valuations.
d) All of the above.

Explanation: Everything above is true. The point of this question is that the Cost Basis when selling an Equity Investment is not necessarily its market value * ownership percentage just before the sale – it could also be affected by acquisitions and sales at different prices, as well as cumulative Net Income and Dividends. Remember that Equity Investments are recorded at historical cost and are not marked to market value.

AFM-03-09: Consolidation Accounting (Noncontrolling Interests), Part 1: Acquiring an Initial Majority Stake (19:27)

In this lesson, you'll get an introduction to Noncontrolling Interests, and you'll learn how to combine and adjust the Balance Sheets when one company acquires a majority stake (between 50% and 100%) of another company.

Table of Contents:

- **5:26:** Part 1: Recording the Starting State
- **10:49:** Part 2: Transaction Assumptions and Goodwill
- **13:19:** Part 3: Assets Side Adjustments
- **15:00:** Part 4: L&E Side Adjustments
- **17:33:** Recap and Summary

Quiz Questions:

13:18 When a company acquires 70% of another company rather than 30%, how does the Goodwill calculation differ?

a) No Goodwill is created in either scenario.
b) Goodwill is created based on 100% of the company's equity purchase price in the first case, but no Goodwill is created in the second case.

c) Goodwill is created based on 70% of the company's equity purchase price in the first case, but no Goodwill is created in the second case.
d) Goodwill is created based on 70% of the company's equity purchase price in the first case, but it's based on 30% in the second case.

Explanation: The second answer choice is correct. Since the statements are consolidated 100%, the new Goodwill must be based on 100% of the target company's equity purchase price when there's a majority-stake acquisition for more than 50% but less than 100% of the target company. In a minority-stake deal, no Goodwill is created because the statements are not consolidated at all. The other answer choices all get this wrong.

AFM-03-10: Consolidation Accounting (Noncontrolling Interests), Part 2: Combining the Financials and Reflecting Net Income and Dividends (20:40)

You'll learn how to consolidate fully the financial statements of the Parent and Sub Co in this lesson, including how the Net Income and Dividends for each entity flow into Common Shareholders' Equity and Noncontrolling Interests.

Table of Contents:

- **0:54:** Part 1: Combined Income Statement
- **5:09:** Part 2: Combined Balance Sheet
- **6:49:** Part 3: Combined Cash Flow Statement
- **12:42:** Part 4: Linking the Balance Sheet
- **18:42:** Recap and Summary

Quiz Questions:

12:42 If a Parent Company owns 80% of Sub Co, and the Sub Co issues a Dividend of $10, what is the *net cash flow impact* on the Parent Company's Cash Flow Statement?

a) It decreases by $8.
b) It decreases by $10.
c) It decreases by $2.
d) It increases by $2.

Explanation: The Parent records 100% of the Sub Co's Dividends, which represents a cash outflow of $10, but then it records 80% * $10 = $8 of Dividends Received, which is a cash inflow. Therefore, cash flow decreases by $2, or $10 * (1 – Ownership Percentage).

AFM-03-11: Consolidation Accounting (Noncontrolling Interests), Part 3: Selling the Stake and Deconsolidating the Financials (20:46)

In this final lesson in the sequence on Noncontrolling Interests, you'll learn how to deconsolidate the financial statements when the parent company sells part of its stake in another company and reduces its ownership to a minority stake.

Table of Contents:

- **1:24:** Part 1: Transaction Assumptions
- **7:29:** Part 2: Balance Sheet Adjustments
- **11:00:** Part 3: Financial Statements in Year 5
- **18:59:** Recap and Summary

Quiz Questions:

10:59 Which of the following does NOT necessarily change on the Balance Sheet when a company moves from a majority stake to a minority stake in another company?

a) The Goodwill that was created in the initial acquisition of the majority stake.
b) The Equity Investments and Noncontrolling Interests.
c) The New Transaction Debt that was used to fund the initial acquisition of the majority stake.
d) Operational Assets, such as Accounts Receivable and Inventory.

Explanation: The third answer choice is correct. When a company goes from a majority stake to a minority stake in another company, the Goodwill created in the original deal must be removed, and Equity Investments and Noncontrolling Interests must change to reflect the new ownership. All of the other company's Assets and Liabilities must also be removed, which means that Operational Assets will definitely change. However, the New Transaction Debt used to fund the initial deal won't necessarily change because the company is not required to repay it when its ownership changes.

AFM-03-12: Stock-Based Compensation, Tax Benefits, and Excess Tax Benefits (U.S. GAAP and IFRS) (21:21)

You'll learn about Stock-Based Compensation and how the exercise of options creates Tax Benefit and Excess Tax Benefit line items on the financial statements in this lesson, as well as how the accounting treatment and Deferred Tax Asset differ under U.S. GAAP and IFRS.

Table of Contents:

- **0:38:** Part 1: Stock-Based Compensation Overview
- **1:47:** Part 2: Assumptions
- **4:12:** Part 3: Linking the Statements
- **9:02:** Part 4: How IFRS is Different
- **15:20:** Part 5: Linking the Statements Under IFRS
- **19:19:** Recap and Summary

Quiz Questions:

9:01 Under U.S. GAAP, if a company records $20 of Stock-Based Compensation on its Income Statement in Year 1, and then its value increases to $40 in Year 5, when employees receive their shares, how does the Deferred Tax Asset change? Assume a 25% tax rate.

a) **It will increase to $5 in Year 1 and decrease back to $0 in Year 5.**
b) It will increase to $5 in Year 1, increase to $10 after that, and then decrease back to $0 in Year 5.
c) It will not change at all because only Excess Tax Benefits affect the DTA.
d) It will increase to $5 in Year and then decrease to ($5) in Year 5.

Explanation: The first answer choice is correct. The DTA under U.S. GAAP reflects only the lack of Cash-Tax Deductibility for the *initial* grant of SBC, and it decreases when that SBC finally becomes deductible for Cash-Tax purposes upon exercise or receipt of the shares. Therefore, the DTA increases by $20 * 25% = $5 in Year and decreases by $5, moving back to $0, in Year 5 in this example. The fact that the SBC's value has doubled does not matter because the Excess Tax Benefits from this increase are recorded directly on the Income Statement and do not affect the DTA on the Balance Sheet.

AFM-03-13: Equity and Debt Investment Types and Unrealized Gains and Losses (Trading or FVPL, AFS or FVOCI, and HTM or Amortized Cost) (17:20)

In this lesson, you'll learn how companies record different types of Equity and Debt securities on their financial statements, and how the treatment for Unrealized Gains and Losses differs based on the security type.

Table of Contents:

- **1:47:** Overview
- **7:08:** Trading or FVPL Securities
- **9:22:** AFS or FVOCI Securities
- **11:02:** HTM or Amortized Cost Securities

- **15:37:** Recap and Summary

Quiz Questions:

9:21 You record a $20 Unrealized Gain on equity securities on the Income Statement. Why does the Deferred Tax Asset decrease?

a) Because you do not actually pay additional Cash Taxes on this Unrealized Gain.
b) Because you will owe Cash Taxes only when these equity securities are actually sold.
c) Because Cash Taxes are less than Book Taxes in this period.
d) All of the above.

Explanation: Everything above is true. Companies do not pay Cash Taxes on Unrealized Gains or save them on Unrealized Losses; they simply change the Deferred Tax Asset such that the actual tax savings happen in some future period when the company finally sells the securities. When there's an Unrealized Gain, Cash Taxes will be less than Book Taxes because the Gain is not taxable, which reduces the DTA. It's the opposite with an Unrealized Loss (Cash Taxes exceed Book Taxes, which increases the DTA), to represent that the real tax savings will come in a future period.

AFM-03-14: Inventory Costs: LIFO vs. FIFO vs. Moving Average Weighted Cost (12:54)

You'll learn how the LIFO, FIFO, and Weighted Average methods for recording Inventory and COGS differ in this lesson, and you'll understand the trade-offs between them and why IFRS does not allow LIFO.

Table of Contents:

- **0:38:** LIFO vs. FIFO
- **3:28:** Excel Demonstration
- **8:58:** Trade-Offs of These Methods
- **11:37:** Recap and Summary

Quiz Questions:

11:37 When would expect the FIFO method for Inventory and COGS to produce a *higher* COGS figure than the LIFO method?

a) When Inventory costs have been rising.
b) When Inventory costs have been falling.

c) When Inventory costs have stayed about the same.
d) When Inventory costs have fluctuated up and down.

Explanation: When Inventory costs have been falling, the most recent Inventory purchased will be less expensive than earlier Inventory purchased. LIFO uses the most recent Inventory, while FIFO uses the oldest Inventory for the costs. Therefore, under FIFO, the COGS shown on the Income Statement will be higher, and under LIFO, the COGS on the Income Statement will be lower. The last two answers do not make sense because you cannot predict anything about COGS under each method in these cases.

AFM-03-15: Pension Accounting, Part 1: Assets, Liabilities, Income Statement, and Cash Flow Statement (U.S. GAAP) (22:06)

In this lesson, you'll learn the fundamentals of pension accounting, including how Pension Assets and Liabilities for defined-benefit plans change, and how the components of the Income Statement and Cash Flow Statement pension line items are created.

Table of Contents:

- **1:32:** What Are Pensions?
- **2:54:** Defined-Contribution Pension Plans
- **4:11:** Defined-Benefit Plans
- **6:03:** Pension Assets
- **8:45:** Pension Liabilities
- **10:41:** Service Cost and Interest Cost
- **14:18:** Income Statement
- **18:18:** Cash Flow Statement
- **20:04:** Recap and Summary

Quiz Questions:

14:17 Which of the following CORRECTLY describes the Pension Service Cost and the Pension Interest Cost?

a) Both are non-cash, non-operational expenses.
b) Both are non-cash, but the Service Cost is operational, while the Interest Cost is not.
c) The Service Cost is a cash, operational expense, while the Interest Cost is non-cash and non-operational.
d) Both are cash, non-operational expense.

Explanation: The second answer choice is correct. Both the Service Cost and Interest Cost are non-cash because they just represent the passage of time or the accrual of future payouts – but no cash payments in the current period. However, the Service Cost is considered operational since it relates to running the plan for employees, while the Interest Cost is considered financial since it's related to the Present Value of future payouts changing.

AFM-03-16: Pension Accounting, Part 2: Links on the Three Statements and IFRS Treatment (16:26)

You'll learn how to link pension-related items in a full 3-statement model in this lesson, including key simplifications, and you'll understand how the accounting treatment of defined-benefit pension plans differs under IFRS.

Table of Contents:

- **0:48:** Pension Links on the Financial Statements
- **8:57:** IFRS Pension Accounting
- **14:04:** Recap and Summary

Quiz Questions:

14:04 Which of the following is NOT a major difference in pension accounting under IFRS vs. U.S. GAAP?

a) Under U.S. GAAP, the difference between Expected and Actual Returns is amortized on the IS; that does not happen under IFRS.
b) The Expected Return on Plan Assets is not an IS line item under IFRS.
c) Service Costs and Interest Costs may be put in different categories on the IS.
d) The tax deductibility of the IS Pension Expense vs. Employer Contributions may differ.

Explanation: The second answer is the correct choice because the Expected Return on Plan Assets *is* still an Income Statement line item under IFRS. It's just that the difference between Expected and Actual Returns is not amortized on the IS, unlike U.S. GAAP. Instead, the difference each year is recorded directly within AOCI in CSE. The last two answer choices are incorrect because they represent *actual differences* between U.S. GAAP and IFRS in the treatment of pensions.